IMO 2026 — Problem 5

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Source: `results/kimi-k3/problem-05/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

solved

Approaches tried

- Tried the first guesses $f \equiv c$ and power functions $f(x) = x^a$. Constants fail immediately by letting $y \to 0^+$ in the right-hand inequality; non-identity powers fail by homogeneity. This was a dead end for the full classification, but it suggested looking for functions making $f(x) + y$ equal to $x + f(y)$.
- Checked $f(x) = x + c$. It works because then $f(x) + y = x + f(y)$, so the middle term is exactly the arithmetic mean of the two numbers x and $f(y)$ occurring in the quadratic and geometric means. This gave the candidate family and shifted the problem to proving that $f(x) - x$ is constant.
- Developed the successful "mean interval" approach (see `approaches/mean-interval.md`): for fixed x, y, the two given inequalities say that $(f(x)+y)^2$ lies in the interval $[(x+f(y))^2 - (x-f(y))^2,\ (x+f(y))^2 + (x-f(y))^2]$. Substituting $x = f(y)$ forces $f(f(y)) = 2f(y) - y$.
- Developed the successful "orbit/forbidden interval" argument (see `approaches/orbit-forbidden-interval` the relation $f(f(t)) = 2f(t) - t$ makes forward iterates arithmetic progressions and positivity forces $f(t) \geq t$. If the increment $f(t) - t$ took two values $c > d$, then the c-orbit eventually falls inside an interval where the left-hand mean inequality with a d-increment point is impossible.

Current best

The complete set of solutions is

$$\boxed{f(x) = x + c \quad (x > 0)}$$

for an arbitrary constant $c \geq 0$. The proof is complete below: setting $M = f(x) + y$, $N = x + f(y)$ rewrites the hypothesis as $|M^2 - N^2| \leq (x - f(y))^2$; taking $x = f(y)$ gives $f(f(y)) = 2f(y) - y$, whose iterates are arithmetic and positivity gives $p(x) := f(x) - x \geq 0$. A descent argument shows that if $p(x) = c > 0$, then the orbit of x under f starts at some $r \in (0, c]$ and is $r + nc$. If p took two values $c > d$, the left-hand inequality would fail on a forbidden interval whose length can be made larger than the spacing c of the c-orbit, forcing a point of that orbit into the forbidden interval. Hence p is constant.

Full proof

We will prove that the solutions are exactly

$$f(x) = x + c \qquad (x > 0)$$

for constants $c \geq 0$.

For fixed $x, y > 0$, put

$$v = f(y), \qquad M = f(x) + y, \qquad N = x + v = x + f(y).$$

Since all quantities are positive, the right-hand given inequality is equivalent to

$$M^2 \geq 4xv.$$

But

$$4xv = (x + v)^2 - (x - v)^2 = N^2 - (x - f(y))^2.$$

Similarly, the left-hand given inequality is equivalent to

$$M^2 \leq 2(x^2 + v^2) = (x + v)^2 + (x - v)^2 = N^2 + (x - f(y))^2.$$

Thus the whole hypothesis is equivalent to

$$\boxed{\left|(f(x) + y)^2 - (x + f(y))^2\right| \leq (x - f(y))^2} \qquad (1)$$

for all $x, y > 0$. This is the key mean estimate recorded in `lemmas/key-mean-estimate.md`.

Now substitute $x = f(y)$ in (1). The right side is 0, so

$$(f(f(y)) + y)^2 = (2f(y))^2.$$

Both sides are positive, hence

$$\boxed{f(f(y)) = 2f(y) - y} \qquad (2)$$

for every $y > 0$.

Let $f^{[0]}(t) = t$ and $f^{[n+1]}(t) = f(f^{[n]}(t))$. Fix t and write

$$p(t) = f(t) - t.$$

Applying (2) repeatedly gives, with $a_n = f^{[n]}(t)$,

$$a_{n+2} = f(f(a_n)) = 2f(a_n) - a_n = 2a_{n+1} - a_n.$$

Since $a_0 = t$ and $a_1 = t + p(t)$, induction yields

$$\boxed{f^{[n]}(t) = t + np(t)} \qquad (n \geq 0). \qquad (3)$$

Every $f^{[n]}(t)$ is positive. If $p(t) < 0$, then $t + np(t) < 0$ for all sufficiently large n, contradicting (3). Therefore

$$\boxed{p(t) = f(t) - t \geq 0} \qquad (t > 0). \qquad (4)$$

Also (3) shows that p is constant on each forward orbit of f; this is `lemmas/iterate-positivity.md`.

We next need a descent statement. Suppose $p(t) = c$ and $t > c > 0$. Put $z = t - c > 0$ and write $e = p(z) \geq 0$. Then

$$f(t) = t + c, \qquad f(z) = z + e = t - c + e.$$

Apply (1) to the pair (t, z). We have

$$f(t) + z = (t + c) + (t - c) = 2t,$$

and

$$t + f(z) = t + (t - c + e) = 2t - c + e,$$

while

$$t - f(z) = t - (t - c + e) = c - e.$$

Hence (1) gives

$$|c - e|\,(4t - c + e) \leq (c - e)^2. \qquad (5)$$

If $c \neq e$, dividing (5) by $|c - e|$ gives

$$4t - c + e \leq |c - e|.$$

But $t > c$, so

$$4t - c + e > 3c + e > |c - e|,$$

where the last inequality is immediate both when $e \geq c$ and when $e < c$. This contradiction proves $e = c$. Consequently

$$f(z) = z + c = t.$$

So, whenever $p(t) = c$ and $t > c$, the point $t - c$ is still positive, has the same increment c, and maps to t; this is `lemmas/descent.md`.

It follows that for any t with $p(t) = c > 0$, repeatedly subtracting c while the current point remains $> c$ is legitimate and preserves the increment c. After finitely many subtractions we reach a number

$$r \in (0, c]$$

with $p(r) = c$. Therefore, by (3),

$$\boxed{f^{[n]}(r) = r + nc \quad (n \geq 0), \qquad p(r + nc) = c.} \qquad (6)$$

We now prove that p is constant on all of $\mathbb{R}_{>0}$; this is the constancy argument in `lemmas/constancy.md`. Suppose not. Then there exist points whose increments are

$$c > d \geq 0.$$

By the previous paragraph, choose $r \in (0, c]$ with $p(r) = c$, and define

$$x_n = r + nc \qquad (n \geq 0),$$

so that $f(x_n) = x_n + c$ by (6).

Choose a point y with $p(y) = d$. If $d > 0$, replace y by a suitable forward iterate $f^{[m]}(y) = y + md$; by (3) this keeps $p(y) = d$ and makes y arbitrarily large. Hence in all cases we may assume that

$$\boxed{2s > c, \qquad \text{where } s := \sqrt{2(c-d)(c+d+2y)}.} \qquad (7)$$

Indeed, if $d > 0$ this is achieved by taking the above iterate large enough; if $d = 0$, then already

$$s^2 = 2c(c + 2y) > 2c^2,$$

so $s > \sqrt{2}\,c$ and certainly $2s > c$.

For this fixed y, we have $f(y) = y + d$. Apply the left-hand part of the original inequality with $x = x_n$:

$$\sqrt{\frac{x_n^2 + (y+d)^2}{2}} \geq \frac{f(x_n) + y}{2} = \frac{x_n + c + y}{2}.$$

Squaring gives

$$2(x_n^2 + (y+d)^2) \geq (x_n + c + y)^2. \qquad (8)$$

Consider the quadratic

$$Q(X) = 2(X^2 + (y+d)^2) - (X + c + y)^2.$$

With $A = c + y$, a direct expansion gives

$$Q(X) = (X - A)^2 - s^2,$$

because

$$s^2 = 2((c+y)^2 - (y+d)^2) = 2(c-d)(c+d+2y).$$

Thus (8) says exactly that x_n must not lie in the open interval

$$I = (A - s, A + s).$$

But $A + s = c + y + s > c \geq r = x_0$, so $x_0 < A + s$. If $x_0 > A - s$, then already $x_0 \in I$, contradicting (8) for $n = 0$. Otherwise $x_0 \leq A - s$. Since $x_n \to \infty$ and the step of the sequence (x_n) is $c < 2s$ by (7), the sequence cannot jump over the interval I: if n is the first index with $x_n > A - s$, then $n \geq 1$, $x_{n-1} \leq A - s$, and

$$x_n = x_{n-1} + c \leq A - s + c < A - s + 2s = A + s.$$

Hence $x_n \in I$, again contradicting (8). Therefore the assumed two increments $c > d$ cannot occur.

So $p(t)$ is a constant $c \geq 0$, i.e.

$$f(t) = t + c \qquad (t > 0).$$

It remains to verify that every such function works. Let $c \geq 0$ and $f(t) = t + c$. For $x, y > 0$, set

$$a = x, \qquad b = f(y) = y + c.$$

Then

$$f(x) + y = x + c + y = a + b.$$

Consequently the required inequalities become

$$\sqrt{\frac{a^2 + b^2}{2}} \geq \frac{a + b}{2} \geq \sqrt{ab}.$$

The right inequality is $(a + b)^2 \geq 4ab$, and the left inequality is $2(a^2 + b^2) \geq (a + b)^2$; both follow from $(a - b)^2 \geq 0$. Hence all functions $f(x) = x + c$ with $c \geq 0$ satisfy the condition, and no other functions do.

Therefore the answer is

$$\boxed{f(x) = x + c \ (x > 0) \text{ for some constant } c \geq 0.}$$